CS & IT ENGINEERING

Discrete Mathematics

Mathematical logic

Lecture No.- 01

By- Satish Sir

Recap of Previous Lecture

Topic Extra Topics in Graph Theory

Topics to be Covered

Topic Logic Part-01

English stm:

Factual.

Conclusion.

$\rightarrow$ Thms. $\rightarrow$ Algo $\rightarrow$ PL

$\downarrow$

S/W

Propositional stmt: → Yes / → NO

Simple.

can not break.

$5>3$ (T)

Compound:

breakable.

$5 \geq 3$

$5>3$ OR $5=3$

connective.

(breakable point)

compound:

S_1 | S_2

Connectives:

→ conjunction.

→ Disjunction.

→ Conditional.

→ Biconditional.

S_1 S_2 | C

conjunction : ($\wedge$) (AND/but)

p	q	$p \wedge q$
T $\wedge$	F $\equiv$	F
F $\wedge$	T $\equiv$	F
F $\wedge$	F $\equiv$	F
T	T	T

compound :

F $\wedge$ ___ $\equiv$ F

F ___ $\wedge$ ___ $\wedge$ ___ $\equiv$ F

*

AND hates false

OR
- → Inclusive OR $(\vee)$ (one, other, both)
- → Exclusive OR (XOR) $(\oplus)$ (one or other)

$\underline{T} \vee \underline{\quad} \equiv T$

OR Loves True.

p	q	$p \vee q$
T	F	T
F	T	T
T	T	T
F	F	F

Conditional stmt ($\rightarrow$)

if p then q.
p implies q
if p, q
q when p
q whenever p
q unless $\neg p$
p only if q
q if p.

$p \rightarrow q$.

Rosen

* if you clear the GATE then she will marry you.

p	q	$p \to q$
→ clear(T)	marry(T)	T
clear(T)	marry(F)	F
clear(F)	marry(T)	T
clear(F)	marry(F)	T

$$\begin{cases} T \to T \equiv T \\ T \to F \equiv F \\ F \to T \equiv T \\ F \to F \equiv T \end{cases}$$

Thm: if $\underbrace{x \text{ is even}}_{T}$ then $\underbrace{x^2 \text{ is also even}}_{F}$.

integer

x is even (T), $x = 2$	x^2 even (T), $x^2 = 4$
x even (T)	x^2 $\cancel{\text{even}}$ (F)
x $\cancel{\text{even}}$ (F), $x = 3$	x^2 even
x even	x^2 even

$T \to T \equiv T$

$T \to F \equiv F \Rightarrow$ Thm is wrong.

$F \to T \equiv T$

$F \to F \equiv T$

$p \to q$

if $\underbrace{\text{clear GATE}}_{p}$ then $\underbrace{\text{she marry you}}_{q}$.

Inverse.

$\neg p \to \neg q$

if you will not clear GATE then she will not marry.

Converse.

$q \to p$

if she marry you then you clear the GATE.

Contrapositive.

$\neg q \to \neg p$.

i.) She not married then you have not cleared.

$p \to q$ is same as $\neg q \to \neg p$

logical equivalent

$p \to q \equiv \neg q \to \neg p$.

planar the $e \le 3n-6$

$p \rightarrow q.$

$p.m \rightarrow$ even

$\neg p \rightarrow \neg q.$

$q \rightarrow p$

$\neg q \rightarrow \neg p.$

if p.m not exist then no. of vertices will not be even

if no. of vertices are not even then p.m will not exist.

p	q	$\neg p$	$\neg q$	A: $p \to q$	B: $\neg q \to \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

logical equivalent

$A \equiv B$

* column of A = column of B.
* A, B are having same behaviour.

$$p \to q \equiv \neg q \to \neg p$$

T   T

F   F

2 mins Summary

Topic One

Topic Two

Topic Three

Topic Four

Topic Five

Logic Part-01

THANK - YOU